An Approximate Newton Method for Saddle Point Problem

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Outline

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Problem Settings

Object function:

$$\min_{x\in\mathbb{R}^n}\max_{y\in\mathbb{R}^m} f(x,y) = \frac{1}{M}\sum_{j=1}^{M} f_j(x,y) \tag{1}$$

where

$$f_j(x) = \frac{1}{|J_j|}\sum_{i\in J_j} l_i(x,y;a_i,b_i)$$

- *Cubic regularized newton method for saddle point models: a global and local convergence analysis.* gives a cubic regularized newton method for the saddle point problem
- *Giant: Globally improved approximate newton method for distributed optimization.* gives a distribute newton method for minimization problem called GIANT
- Motivated by the above two works, we propose an approximate newton method for solving (1)

Notations

- $z = (x; y)$
- $F(z) = \begin{pmatrix} \nabla_x f(x,y) \\ -\nabla_y f(x,y) \end{pmatrix}.$
- $\nabla F(z) = \begin{pmatrix} H_{xx} & H_{xy} \\ -H_{xy}^T & -H_{yy} \end{pmatrix}$
- $F_i(z)$, $\nabla F_i(z)$, $H_{xx,i}, \cdots$ for the local object function $f_i(z)$
- Merit function: $m(z) := \frac{1}{2}\|F(z)\|^2$
- Sub-problem :

$$\begin{aligned} \min_{x\in\mathbb{R}^n} \max_{y\in\mathbb{R}^m} f_i^k(x,y;\gamma^k) \quad &= f(z^k) + \langle g^k, z - z^k\rangle + \tfrac{1}{2}(z-z^k)^\top H_i^k (z-z^k) \\ &+ \tfrac{\gamma^k}{3}\|x-x^k\|^3 - \tfrac{\gamma^k}{3}\|y-y^k\|^3 \end{aligned} \tag{2}$$

Assumptions

- f_j is strongly convex in x and strongly concave in y with modulus μ.
- $\|F(z) - F(z')\| \leq L\|z - z'\|, \quad \forall z, z'$..
- $\|\nabla F(z) - \nabla F(z')\| \leq L_2\|z - z'\|, \quad \forall z, z'$.
- the level set $\mathcal{Z} := \{z : m(z) \leq m(z^0)\}$ and denote $D := \max\{\|z - z^0\| : z \in \mathcal{Z}\}$.
- The approximate Hessian holds that $\|H_i - H\|_2 \leq \eta\|H\|_2$.

D-CRN-SPP:

Require: Input ϵ, $\bar{\gamma} > 0$,machine num :M; $\rho, \alpha \in (0,1)$, f,flag=0

while $m(z_k) > \epsilon$ **do**

  for $i \in \{1, 2, \cdots, M\}$ **in parallel over computers do**

    Calculate g_i^k

  $g^k = \frac{1}{M}\sum_{i=1}^{M} g_i^k$

  $\gamma_i^k \leftarrow \bar{\gamma}$

  while true do

    for $i \in \{1, 2, \cdots, M\}$ **in parallel over computers do**

      $(\tilde{x_i}^{k+1}, \tilde{y_i}^{k+1}) = \arg\min_x \max_y f_{Ji}^k(x, y; \gamma_i^k)$

      $d_i^{k+1} = (x_i^{k+1} - x^k, y_i^{k+1} - y^k)^t$

      if $\gamma_i^k(\|u_i^k\| + \|v_i^k\|) > \mu$ **then**

        $\gamma_i^k \leftarrow \rho \cdot \gamma_i^k$

      else break

    $\gamma^k = min_i(\gamma_i^k)$

    for $i \in \{1, 2, \cdots, M\}$ **in parallel over computers do**

      Call local solver :

      $(\tilde{x_i}^{k+1}, \tilde{y_i}^{k+1}) = \arg\min_x \max_y f_{Ji}^k(x, y; \gamma^k)$

      $d_i^{k+1} = (x_i^{k+1} - x^k, y_i^{k+1} - y^k)^t$

  Denote $d^k \leftarrow \frac{\Sigma_{i=1}^{M} d_i^k}{M}$

  if $m(z^k) < \frac{\mu^5}{8LL_2^2}$ **then**

    flag=1

  if flag=0 **then**

    $z^{k+1} \leftarrow z^k + \alpha d^k$

  else if flag=1 **then**

    $z^{k+1} \leftarrow z^k + d^k$

  $k \leftarrow k + 1$

return z^k

Contributions

In Algorithm 1, we present a distributed optimization algorithm for solving the saddle point problem.

- The first applicable distribute second-order method for the saddle point problem.
- Establish the convergence analysis for both local and global in general case of for approximate second order method.
- The local convergence rate is better than changing step-size way in [**1**].

The relation between Distance to the saddle point and the Merit fucntion

Theorem 1

$$\frac{2}{L^2} m(z) \leq \|z - z^*\|^2 \leq \frac{2L}{\mu^3} m(z)$$

- This theorem is easy to build on the work of [**1**].
- Any small $\|z - z^*\|$ can achieved by a small enough $m(z)$ which is the key for the step size choice and the local convergence.
- We want $\|z - z^*\| \leq \frac{\mu}{2L_2}$ in the local analysis. So when implement the algorithm, we only need to check if $m(z) \leq \frac{\mu^5}{8LL_2^2}$.

Global Convergence

Theorem 2 (Global Iteration Complexity)

Let $\{z^k\}$ be generated by Algorithm 1, with $\alpha = \frac{\mu^2}{2L_m}$ and $L_m = L^2 + LL_2D$, and the step size always be α, then the sequence $\{m(z^k)\}$ converges linearly to 0, More specifically, when $\eta = \min\left\{\frac{2}{\sqrt{9(L+1)^2-12L}+3(L+1)}, \frac{\mu^2}{4L_2}, \frac{1}{2}\right\}$ we have

$$m(z^{k+1}) \leq \left(1 - \frac{\mu^2}{14L_m}\right)^2 m(z^k) \tag{3}$$

- This theorem is based on the Theorem 3.6 in [1] which gives the convergence result: $m(z^{k+1}) \leq \left(1 - \frac{\mu^2}{6L_m}\right)^2 m(z^k)$.
- Comparing above 2 iteration complexities, the approximate algorithm converges slower than the precise algorithm, but if η is small enough, they have the same convergence rate as :$\mathcal{O}\left(\left(\kappa^2 + \kappa \cdot \frac{L_2}{\mu}\right)\ln\left(\frac{1}{\epsilon}\right)\right)$.

Key Lemma and the proof details

Recall the sub problem for each local machine:

$$\min_{x\in\mathbb{R}^n}\max_{y\in\mathbb{R}^m} f_i^k(x,y;\gamma^k) \quad = f(z^k)+\langle g^k, z-z^k\rangle+\tfrac{1}{2}(z-z^k)^\top H_i^k(z-z^k)$$
$$+\tfrac{\gamma^k}{3}\|x-x^k\|^3-\tfrac{\gamma^k}{3}\|y-y^k\|^3$$

1. The stationary condition for the local solver:

$$\nabla F_i(z^k)d_i^k = \begin{pmatrix} H_{xx,i}^k u_i^k + H_{xy,i}^k v_i^k \\ -(H_{xy,i}^k)^\top u_i^k - H_{yy,i}^k v_i^k \end{pmatrix} = -F(z^k) - \gamma^k \begin{pmatrix} \|u_i^k\| u_i^k \\ \|v_i^k\| v_i^k \end{pmatrix}. \quad (4)$$

2. Gradient related direction

$$\langle \nabla m(z^k), d_i^k\rangle \le -\frac{\mu^2}{2}\|d_i^k\|^2 + (g^k)^T\Delta(H_i^k)d_i$$
$$(g^k)^T\Delta(H_i^k)d_i \le \frac{\eta L}{2}(\|d_i^k\|^2 + LD\|F(z^k)\|)$$

Key Lemma and the proof details

3. Sufficient Descent in $m(z^k)$

$$m(z^{k+1}) - m(z^k) \leq -\frac{\mu^4}{8L_m M}\sum_i \|d_i^k\|^2 + \frac{\alpha\sum_i (g^k)^T \Delta(H^k) d_i}{M} \tag{5}$$

Where α is the proper step size we choose to guarantee the sufficient descent. $\alpha = \frac{\mu^2}{2L_m}$

4. The relation between $\|F(z^{k+1})\|$ and $\|F(z^k)\|$

$$\|F(z^{k+1})\| \leq (1-\alpha+\tfrac{\eta}{1-\eta}\alpha)\|F(z^k)\| + (\tfrac{1}{1-\eta}\bar{\gamma}\alpha + \tfrac{\alpha^2 L_2}{2})\frac{\sum_i \|d_i^k\|^2}{M} \tag{6}$$

Local Convergence

Theorem 3 (Local Convergence)

Let $\{z^k\}$ be generated by Algorithm1 with $\bar{\gamma} = \frac{L_2\mu^2}{2L^2}$, then there exists a constant $K > 0$ such that for all $k \geq K$ we have:

$$\|z^{k+1} - z^*\| \leq \max\left\{\frac{2L_2}{\mu}\|z^k - z^*\|^2, \frac{2\eta L}{\mu}\|z^k - z^*\|\right\} \tag{7}$$

- This Theorem is based on the Theorem 3.6 in [**1**] which gives the iteration complexity : $\|z^{k+1} - z^*\| \leq \frac{LL_2}{\mu^2}\|z^k - z^*\|^2$
- Noted that without the step size change method proposed in our Algorithm, the local convergence can only achieves

$$\|\hat{z}^{k+1} - z^*\| \leq \max\left\{\frac{2L_2L}{\mu^2}\|z^k - z^*\|^2, \frac{2\eta L^2}{\mu^2}\|z^k - z^*\|\right\} \tag{8}$$

AUC Problem

We consider the following AUC maximization problem which can be written in the form (in [3]) of:

$$\min_{\mathbf{x}\in\mathbb{R}^{d+2}} \max_{\mathbf{y}\in\mathbb{R}} f(\mathbf{x}, y) \triangleq \frac{1}{N}\sum_{i=1}^{N} f_i(\mathbf{x}, y; \mathbf{a}_i, b_i, \lambda)$$

where $\mathbf{x} = [\mathbf{w}; u; v]^T$,

$$\begin{aligned} f_i(\mathbf{x}, y; \mathbf{a}_i, b_i, \lambda) =& \frac{\lambda}{2}\|\mathbf{x}\|_2^2 - p(1-p)y^2 \\ &+(1-p)((\mathbf{w}^T\mathbf{a}_i - u)^2 - 2(1+y)\mathbf{w}^T\mathbf{a}_i)\mathbb{I}_{b_i=1} \\ &+p((\mathbf{w}^T\mathbf{a}_i - v)^2 + 2(1+y)\mathbf{w}^T\mathbf{a}_i)\mathbb{I}_{b_i=1} \end{aligned}$$

AUC Problem

We give the method to fit the approximate assumptions. To find the small enough η for $\|H_i - H\|_2 \leq \eta \|H\|_2$.

1. Noted that $\mathbf{x} \in \mathbb{R}^{d+2}$ and $y \in \mathbb{R}$, we will use approximate $H_{\mathbf{xx}}$ and precise $H_{\mathbf{x}y}, H_{yy}$ in the distribute algorithm. and thus we only need to find η for $\|H_{\mathbf{xx},i} - H_{\mathbf{xx}}\|_2 \leq \eta \|H_{\mathbf{xx}}\|_2$
2. Calculate the $H_{\mathbf{xx}}$ and $H_{\mathbf{xx},i}$ They can be written as:

$$H_{\mathbf{xx}} = \lambda I_{d+2} + X^\top X \tag{9}$$

$$H_{\mathbf{xx},i} = \lambda I_{d+2} + X^\top \mathbf{S}_i \mathbf{S}_i^\top X \tag{10}$$

where $\mathbf{S}_i$ is the sketching matrix and:

$$X = [\frac{1}{\sqrt{n}}\bar{\mathbf{x}}_1^\top; \frac{1}{\sqrt{n}}\bar{\mathbf{x}}_2^\top; \cdots; \frac{1}{\sqrt{n}}\bar{\mathbf{x}}_n^\top] \in \mathbb{R}^{n\times(d+2)} \tag{11}$$

$$\bar{\mathbf{x}}_i = \sqrt{2(1-p)}[\mathbf{a}_i; -1; 0]^T \mathbb{I}_{b_i=1} + \sqrt{2p}[\mathbf{a}_i; 0; -1]^\top \mathbb{I}_{b_i=-1} \tag{12}$$

Sketching Matrix

$X \in \mathbb{R}^{n\times(d+2)}$, $C = \mathbf{S}^T X \in \mathbb{R}^{s\times(d+2)}$ as *sketch* of X.
The *sketching matrix* $\mathbf{S} \in \mathbb{R}^{n\times s}$ (typically $s \ll n$) make the rows of C be made up of a randomly selected subset of the rows of X.

$$\mathbb{P}\left(\mathbf{c}_i = \mathbf{x}_j/\sqrt{sp_j}\right) = p_j, \quad \text{for all } \ j = 1, 2, \ldots, n,$$

where $\mathbf{c}_i$ and $\mathbf{x}_j$ are i^{th} and j^{th} rows of $\mathbf{C}$ and X

Lemma 4

Let $\eta, \delta \in (0,1)$ be fixed paramters. Let $U \in R^{n\times\rho}$ be any fixed matrix with orthonormal columns. Let $S_1, \cdots, S_m \in \mathbb{R}^{n\times s}$ be independent uniform sampling matrices with $s \geq \frac{3\mu\rho}{\eta^2}\log\frac{\rho m}{\delta}$,It holds with probability at least 1-δ that:

$$\left\|U^T S_i S_i^T U - I_\rho\right\|_2 \leq \eta \quad for\ all\ i \in [m]$$

AUC Problem

Using the **Matrix Sketching** technique and the lemma 4 we have the following Theorem 5 when solving AUC problem.

Theorem 5

let m be the number of partitions, let μ be the row coherence of X given in (11). For a given η, σ, assume that the local sample size satisfy that $s \geq \frac{3\mu(d+2)}{\eta^2} log \frac{m(d+2)}{\sigma}$, It holds with probability $1-\sigma$ that :

$$\|H_i - H\|_2 \leq \eta \|H\|_2 \tag{13}$$

AUC Problem

Combining Theorem 3 and Theorem 5, and note that $L_2 = 0$, when n is at least $3\mu(d+2)m\log(m(d+2))$ the local convergence rate for AUC:

$$\|z^{k+1} - z^*\| \leq 2\sqrt{\frac{3\mu(d+2)mlog(m(d+2)/\delta)}{n}}\kappa\|z^k - z^*\| \tag{14}$$

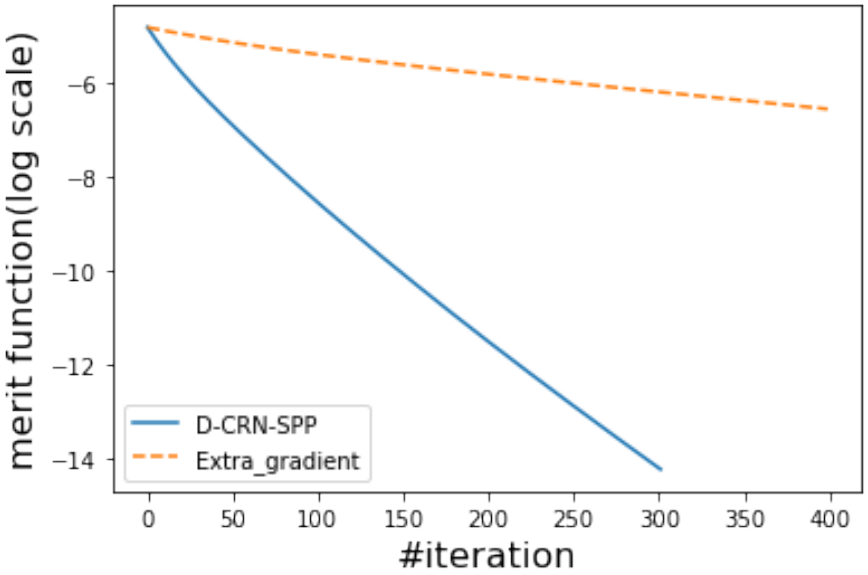
Table: The iteration complexity for AUC

Methods	Iterations
Extra Gradient	$\mathcal{O}\left(\kappa log(\frac{1}{\epsilon})\right)$
Distribute-CRN-SPP	$\mathcal{O}\left(\frac{log(1/\epsilon)}{log(n/\mu dm\kappa^2)}\right)$

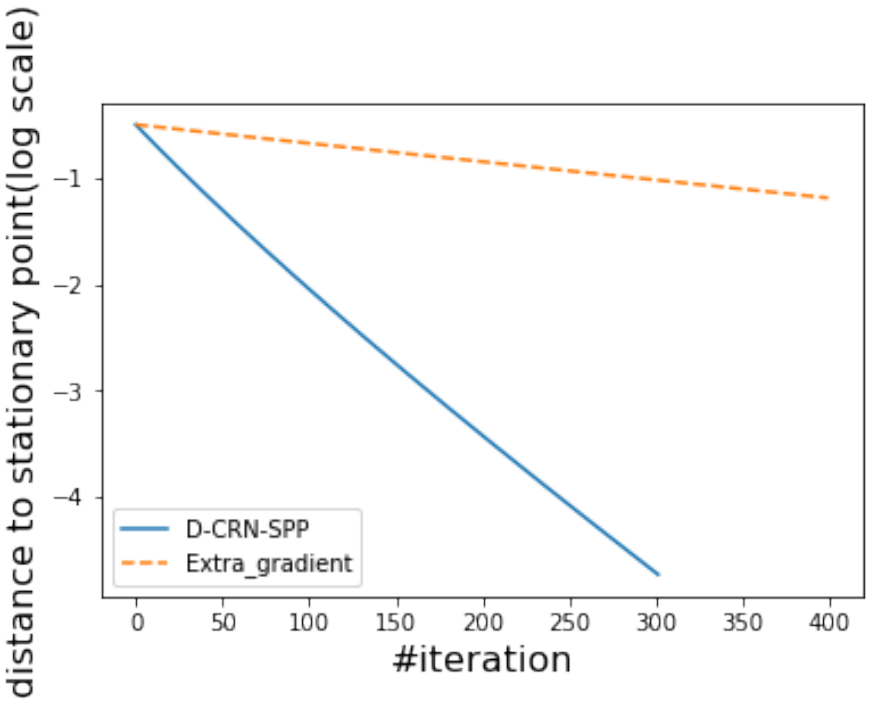
Noted that with n large enough (i.e $n = \mathcal{O}\left(\kappa^3 dm\mu\right)$). Our Algorithm is much better than Extra Gradient when solving AUC problem.

AUC problem

Data set: cov.libsvm.binary; n=581,012, d=54



(a) Merit function



(b) Distance to saddle point

Experiment with object function in [1]

Here we do the experiment with object function in [**1**] which shows that for some cases, D-CRN-SPP and CRN-SPP may enjoys almost the same convergence rate.

$$\begin{aligned} \min_{x\in\mathbb{R}^n}\max_{y\in\mathbb{R}^m} f(x,y) \quad = \quad & \frac{1}{M_1}\sum_{i=1}^{M_1}\ln(1+e^{-a_i^\top x})+\frac{1}{2}\|x\|^2 \\ & +x^\top Ay-\frac{1}{M_2}\sum_{j=1}^{M_2}\ln(1+e^{-b_j^\top y})-\frac{1}{2}\|y\|^2, \end{aligned} \tag{15}$$

Experiment with object function in [1]

We set n=100, m=200, M1=M2=1000

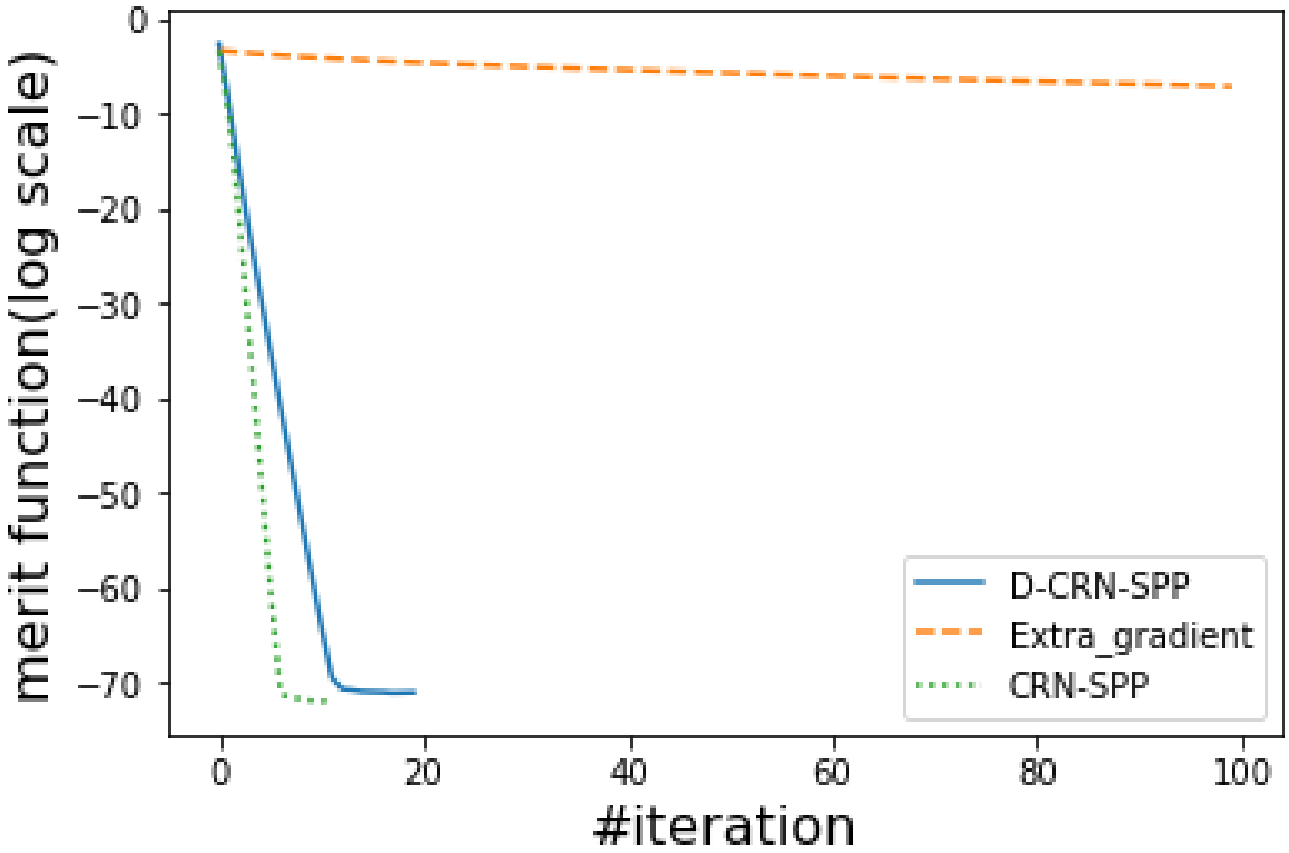


Figure: Merit function via iteration

Future Works

- Can we find out a theory like Theorem 5 in a general case for the saddle point matrix approximation? $\|H_i - H\|_2 \leq \eta \|H\|_2$
- Is there other way to approximate the saddle point matrix? For example, can we use BFGS update formula to solve saddle point problem? (We tried in May, but have not figured out a good way yet)

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